

Multi-Objective Optimization

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1 Intro

Why multi-objective?

In reality, normally, there are more than one objective when making decisions:

- To minimize cost
- To minimize completion time
- To minimize changes in the number of human resources
- To maximize performance

In many cases, these objectives are conflicting:

- When minimizing completion time, cost normally grows
- When minimizing cost, performance/production/sales is normally sacrificed
- **It is not possible to optimize all of them at the same time**

2 MCDM, MADM and MODM

2.1 MCDM (multiple criteria decision making)

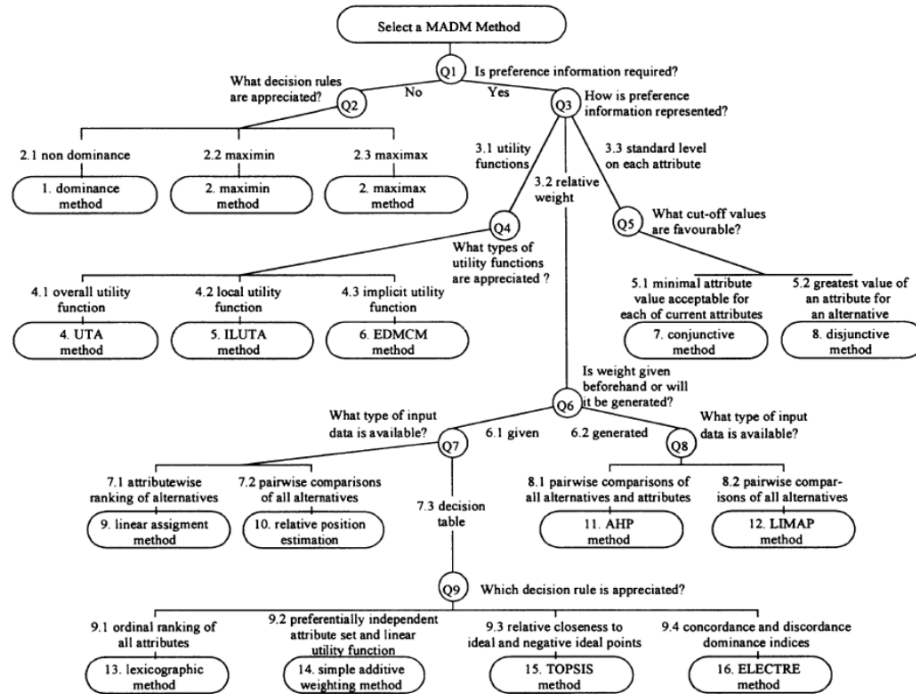
2.1.1 MODM (multiple objective DM)

- Normally continuous and uncountable set of alternatives/solutions
- Aims to generate solutions

2.1.2 MADM (multiple attribute DM)

- Discrete and limited set of alternatives/solutions
- Aims to rank existing alternatives/solutions

Selecting a MADM method



2.2 What is a multi-objective optimization model?

Same as a normal optimization model, but with multiple objectives

2.3 Which solution is the best?

- Decision variables: how much goods to be sent on each mode of transport:
 - Road
 - Rail
- Objectives:
 - Minimization of total cost
 - Minimization of average delivery time

Setup:

Imagine we have three alternatives within the region in the feasible solution set:

- A: Cost = 60, delivery time = 8
- B: Cost = 40, delivery time = 14
- C: Cost = 30, delivery time = 24

2.4 Possible approach

To merge multiple objectives into a single objective

- Minimize Combined Objective (with delivery time measured in terms of customer dissatisfaction cost)
- Assume that in the example given on the previous page, Combined Objective is simply the sum of cost and delivery time

If objectives are equally important, which solution is the best?

- A: Combined Objective = 54
- B: Combined Objective = 54
- C: Combined Objective = 68

Is it really that easy?

- Are all objectives equally important?
- If objectives are not equally important, how could one specify their relative importance (weight)?
- The decision maker has to specify the relative importance of the objectives “a priori”, i.e., in advance
 - Is cost more important than delivery time?
 - If yes, how much? Twice as important? 5 times as important?
- This technique turns the multi-objective optimization problems into a single-objective one
- This approach belongs to so-called “preference-based procedures”
- It is called Weighted Sum Method

3 Weighted Sum Method

- The multi-objective model turns into a single-objective one as follows:

$$\begin{aligned}\min Z &= \sum_{i=1}^m w_i f_i(x_1, x_2, \dots, x_n) \\ &\text{s.t.} \\ g_1(x_1, x_2, \dots, x_n) &\leq b_1 \\ g_2(x_1, x_2, \dots, x_n) &\leq b_2 \\ &\vdots \\ g_k(x_1, x_2, \dots, x_n) &\leq b_k\end{aligned}$$

- The decision maker must specify the values of the weights $w_1, w_2, \dots, w_m$ in advance
- Objectives must be normalized as well, i.e., must have the same scale
- Weights can be determined by pairwise comparisons
- By involving the decision maker in the process of optimization, weights can be changed interactively

4 ϵ -constrained method

- As in the weighted sum method, the multi-objective model turns into a single-objective one
- One objective is chosen as the main objective to be optimized
- Other objectives will appear in the model as constraints
 - In the case of minimization, a maximum acceptable level must be set for these objectives

$$\begin{aligned} \min Z &= f_{\beta}(x_1, x_2, \dots, x_n) \\ \text{s.t.} \\ f_i(x_1, x_2, \dots, x_n) &\leq \epsilon_i \quad \forall i = 1, \dots, m, i \neq \beta \\ g_j(x_1, x_2, \dots, x_n) &\leq b_j \quad \forall j = 1, \dots, k \end{aligned}$$

- The decision maker must specify the values of the constraint limits $\epsilon_1, \dots, \epsilon_m$ in advance
- As in the weighted sum method, by involving the decision maker in the process of optimization, limits can be changed interactively

4.1 Issues

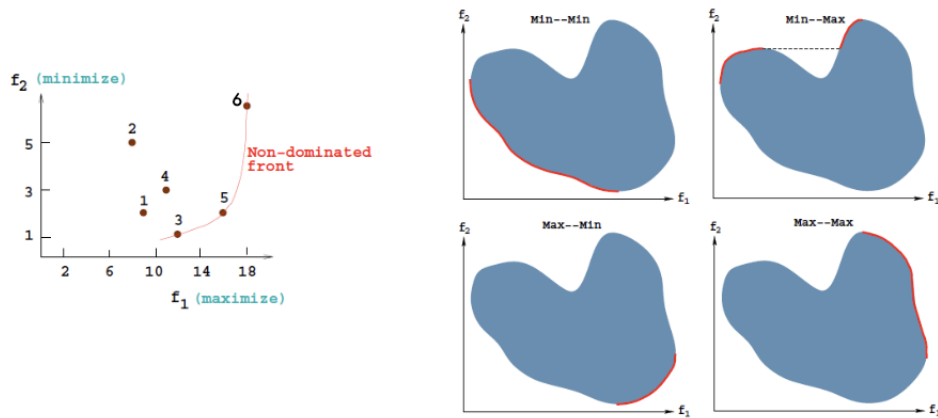
- What if the objectives are not comparable?
- What if the decision maker does not know their relative importance in advance?
- What if one cannot set limits for the objectives?
- Which objective to be optimized and which ones should get turned into constraints?
- What if the decision maker wants to have different solutions in hand, observe the trade-offs between the objectives, and then decide which solution to choose?
- All of the aforementioned methods require higher-level and a-priori information
- If it is so easy to turn a multi-objective problem into a single-objective one, why should one go for multi objectives at all?

5 Domination and Pareto Optimality

- If solution A is better than or equal to solution B in all of the objectives and strictly better than B in at least one objective, then solution A dominates solution B
 - Example: We want to buy a car, and the objectives are to minimize emissions and total cost ownership. The alternatives are a Skoda, Mercedes and a Tesla. Tesla has the lowest emissions, but is the most expensive as well. The Skoda is the cheapest, and has more emissions than the Tesla, but less than the Mercedes.
 - The Skoda dominates the Mercedes, while the Tesla cannot do the same. However, neither the Skoda nor the Tesla can dominate each other
- Given a set of P solutions, the set $P' \subset P$ is called the Non-Dominated set, when P' is the set of all solutions that cannot be dominated by any other solution in P
 - The non-dominated set in the example above consists of the Skoda and the Tesla
- **Pareto-Optimal set** is the set of feasible solutions which cannot be dominated by any other feasible solution. Each member of the Pareto-optimal set is called a **Pareto-optimal solution**
 - In other words, Pareto-optimal set is P' , when P is the set of all feasible solutions (feasible set)
- So we are NOT talking about ONE optimal solution any longer
- The weighted sum and ϵ -constraint methods do not necessarily generate a Pareto-optimal solution
- These classical methods generate only one optimal solution. It would be good if we can generate all of the Pareto-optimal solutions, or at least a countable and finite set of uniformly distributed Pareto-optimal solutions.

5.1 Pareto-optimal and non-dominated fronts

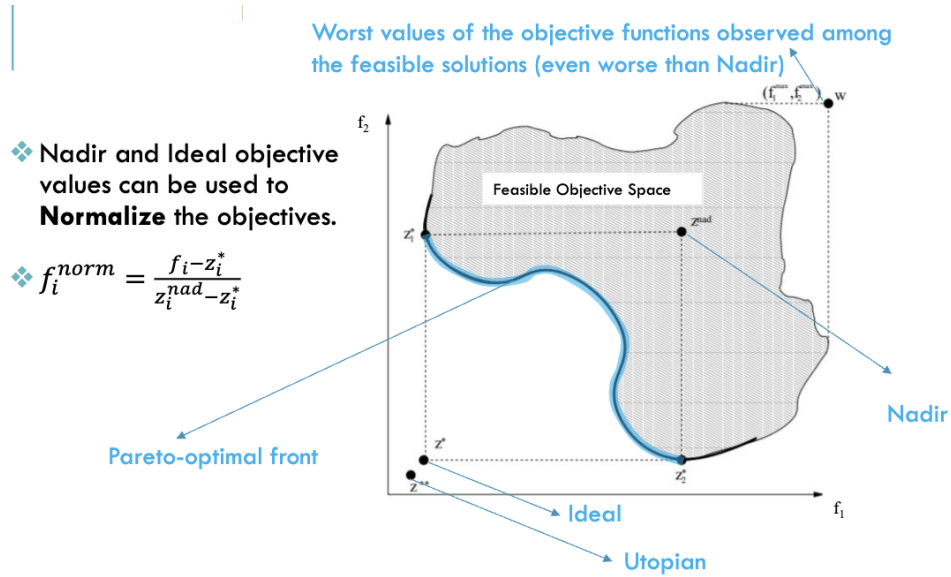
Mapping of Pareto-optimal and non-dominated sets into objective space: The figure on the right shows the Pareto-optimal front



5.2 Some Definitions

- If one optimizes the objectives separately and puts the optimal values of the objectives into a vector, that will be called **the ideal objective vector**
 - If the objectives are conflicting, no feasible solution can generate all of the objective values in the ideal objective vector at the same time. Therefore, the ideal objective vector does not represent a feasible solution
- A vector of objective values whose elements are strictly better than the corresponding elements of the ideal objective vector (and therefore better than all of the feasible solutions), is called a **utopian objective vector**
 - Some algorithms require a utopian objective vector to perform
- A vector of objective values whose elements are the worst values of the objectives observed in the Pareto-optimal set is called **the Nadir objective vector**
 - The Nadir objective vector may or may not represent a feasible solution

5.2.1 Ideal, Utopian and Nadir objective vectors



6 Ideal multi-objective optimization

Process

1. Find a set of all Pareto-optimal solutions
2. Choose one solution from the set (higher-level information applied a-posteriori)

Challenges

- Converging to the Pareto-optimal front
- Finding a diversely and evenly distributed set of Pareto-optimal solutions

7 A simple method to estimate Pareto-optimal front

Assume you have two objectives f_1 and f_2

1. Choose an objective, say f_1 , and determine its optimal value V_1 . For the same solution that gives you V_1 , find the value of f_2 and denote it as V_2 . Then (V_1, V_2) is a point on the Pareto-optimal front.
2. For values V of f_2 that are better than V_2 , solve the optimization problem in step 1 with the additional constraint that the value of f_2 is at least as good as V . Varying V (over values of V preferred to V_2) yields other points on the Pareto-optimal front
3. Step 1 located one endpoint on the Pareto-optimal front. Now determine the best value of f_2 that can be attained, to obtain the other endpoint of the Pareto-optimal front

7.1 General form of the previous algorithm

- Assume we know a classical optimization method to optimize a single-objective optimization problem P with constraints $(\vec{X} \in S)$ and can find a near-optimum solution $\vec{x}^*$. Assume we want to find K different efficient solutions
- Step 1: Find individual optimum solutions $\vec{x}^{*i}$ for all objectives, where $i = 1, 2, \dots, M$
- Step 2: Choose K points $\epsilon^{(k)}$ uniformly in the $(M - 1)$ -dimensional plane having objectives $i = 1$ to $i = M - 1$ as coordinate directions
- Step 3: Solve each subproblem ($k = 1, \dots, K$) as follows:

$$\begin{aligned} \min Z &= f_M(\vec{x}) \\ \text{s.t.} \\ f_i(\vec{x}) &\leq \epsilon_i^{(k)}, \quad \forall i \in \{1, 2, \dots, M - 1\} \\ \vec{x} &\in S \end{aligned}$$

Call the optimal solution $\vec{x}^{*(k)}$ and corresponding objective vector $f^{*(k)}$

- Step 4: Declare the non-dominated set as the set of efficient sets:

$$F = \text{non-dominated}(f^{*(1)}, \dots, f^{*(k)})$$

8 A simple linear multi-objective optimization example

Cement	Type1	Type2
Profit /ton (USD)	700	900
CO2 emissions /ton (g)	2100	2900
Mixer hrs /ton	3	4
Soil /ton (tons)	1.5	1.2

Desirable profit /week (USD)	50,000
Desirable CO2 emissions /week (g)	140,000
Mixer hrs available /week	324
Soil available /week (tons)	150

Let:

- x_i : Tons of cement type i produced per week

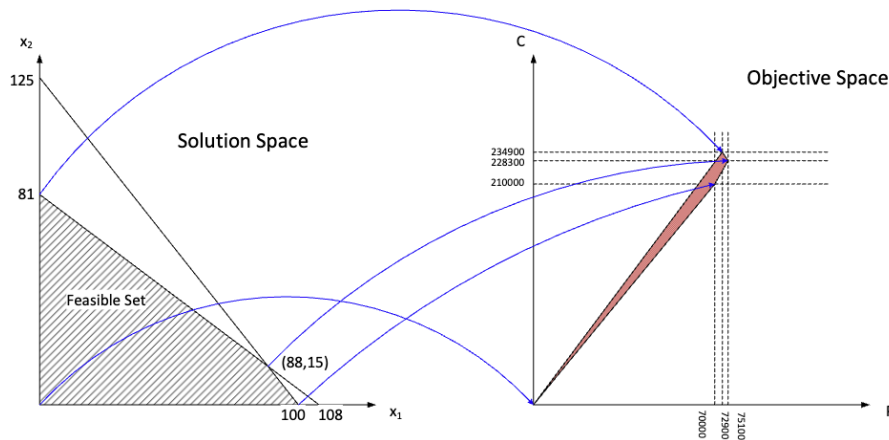
8.1 Version 1: profit maximization

$$\begin{aligned}
 \max Z &= 700x_1 + 900x_2 \\
 \text{s.t.} \\
 3x_1 + 4x_2 &\leq 324 \\
 1.5x_1 + 1.2x_2 &\leq 150 \\
 x_i &\geq 0 \quad \forall i \in \{1, 2\}
 \end{aligned}$$

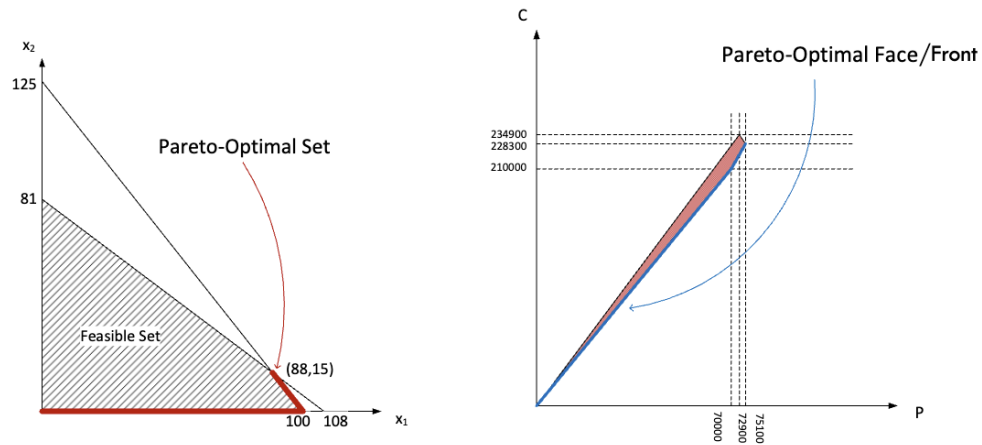
8.2 Version 2: CO2 emissions minimization

$$\begin{aligned}
 \min Z &= 2100x_1 + 2900x_2 \\
 \text{s.t.} \\
 3x_1 + 4x_2 &\leq 324 \\
 1.5x_1 + 1.2x_2 &\leq 150 \\
 x_i &\geq 0 \quad \forall i \in \{1, 2\}
 \end{aligned}$$

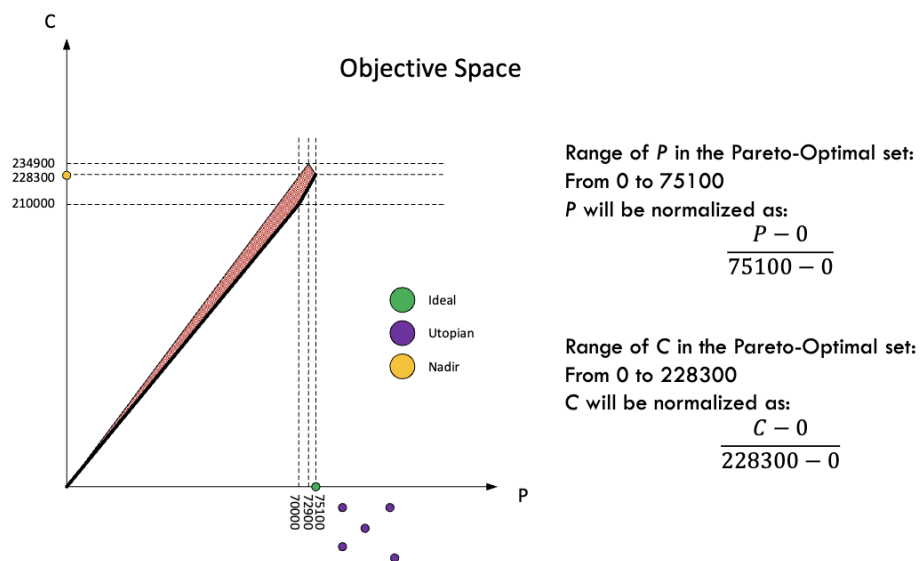
Solution space and objective space



Pareto-optimality



Ideal, Utopian and Nadir solutions



8.3 Version 3: Maximize weighted sum of (normalized) profit and (negative) CO2 emission

Assume profit is as important as CO2 emissions

$$\begin{aligned}\max Z &= \frac{1}{75,100}(700x_1 + 900x_2) - \frac{1}{228,300}(2100x_1 + 2900x_2) \\ \text{s.t.} \\ 3x_1 + 4x_2 &\leq 324 \\ 1.5x_1 + 1.2x_2 &\leq 150 \\ x_i &\geq 0 \quad \forall i \in \{1, 2\}\end{aligned}$$

(the 75k comes from the max profit, and the 228k is from the co2 emission in version 1 where we optimized for profit maximization)

8.4 An easier way of normalizing objective functions

Consider one of the objective functions like F

1. Put aside all other objective functions and only keep F
2. Maximize F with the same set of constraints as the main problem, name it $F_{\max}$
3. Minimize with the same set of constraints as the main problem, name it $F_{\min}$
4. The normalized value of F will be $\frac{F - F_{\min}}{F_{\max} - F_{\min}}$

If $F_{\min} = 0$, then simply you have to multiply $\frac{1}{F_{\max}}$ to normalize it

In our example, you have to multiply $\frac{1}{75,100}$ into profit and $\frac{1}{234,900}$ into CO2 emissions to normalize them

8.5 Version 4: maximize profit s.t. CO2 emission constraint (ϵ -constrained)

Assume the maximum acceptable CO2 emission is 140,000

$$\begin{aligned}\max Z &= 700x_1 + 900x_2 \\ \text{s.t.} \\ 3x_1 + 4x_2 &\leq 324 \\ 1.5x_1 + 1.2x_2 &\leq 150 \\ 2100x_1 + 1900x_2 &\leq 140000 \\ x_i &\geq 0 \quad \forall i \in \{1, 2\}\end{aligned}$$

9 Goal programming

Minimize the total **absolute** deviation of the objectives from the given **goals**

$$\begin{aligned}
 \min Z &= \sum_{i=1}^m w_i |f_i(x_1, x_2, \dots, x_n) - \text{Goal}_i| \\
 &\text{s.t.} \\
 &g_1(x_1, x_2, \dots, x_n) \leq b_1 \\
 &g_2(x_1, x_2, \dots, x_n) \leq b_2 \\
 &\vdots \\
 &g_k(x_1, x_2, \dots, x_n) \leq b_k
 \end{aligned}$$

9.1 Linearizing absolute deviations

$$\begin{aligned}
 \min Z &= \sum_{i=1}^m w_i (F_i^+ + F_i^-) \\
 &\text{s.t.} \\
 F_i^+ - F_i^- &= f_i(x_1, \dots, x_n) - \text{Goal}_i \quad \forall i = 1, \dots, m \\
 F_i^+ \cdot F_i^- &= 0 \quad \forall i = 1, \dots, m \\
 F_i^+, F_i^- &\geq 0 \quad \forall i = 1, \dots, m \\
 g_j(x_1, x_2, \dots, x_n) &\leq b_j \quad \forall j = 1, \dots, k
 \end{aligned}$$